Spectral Error Indicators for Neural PDE Solvers

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Abstract

Physics-Informed Neural Networks (PINNs) act as mesh-free solvers for partial differential equations (PDEs), yet verifying their accuracy remains challenging. Standard error indicators depend on PDE residual magnitude, which poorly correlates with true absolute error in stiff flows, masking large localized inaccuracies. We propose a novel post-hoc spectral error indicator framework addressing this limitation. By leveraging Dynamic Mode Decomposition (DMD) on intermediate PINN training snapshots, we capture the spectral convergence dynamics of the network. A Convolutional Neural Network (CNN) maps these localized spatial DMD patches to true spatial errors. Our validation on stiff 1D Allen-Cahn and 2D Navier-Stokes equations shows near-perfect correlation ($r > 0.99$) during in-distribution cross-validation, substantially outperforming standard residual methods. To enable zero-shot generalization, we introduce a deterministic Pointwise Temporal Variance metric. Rigorous multi-seed testing reveals that while Temporal Variance outperforms spatial CNNs for parameter interpolation on Kovasznay flow, it exhibits substantial stochastic noise ($r = 0.56 \pm 0.18$). This confirms that extracting stable zero-shot indicators from chaotic training dynamics remains a fundamental challenge. Nevertheless, this approach provides a baseline solver-agnostic targeting system that outperforms standard residuals, unlocking computational savings for parametric sweeps.

1 Introduction

Physics-Informed Neural Networks (PINNs) integrate physical governing equations into the loss function of deep neural networks, enabling the mesh-free solution of PDEs. Despite their success, verifying the accuracy of a PINN solution remains a significant challenge. Unlike classical methods (e.g., Finite Element Method) which offer rigorous a posteriori error bounds, PINNs lack reliable, computationally cheap error certification mechanisms. A PINN may smoothly satisfy boundary conditions while harboring massive localized errors near shock fronts or sharp phase interfaces.

The ultimate goal of a neural PDE error indicator is to enable massive, automated parameter sweeps (e.g., varying aerodynamic shapes or Reynolds numbers) without requiring expensive, high-fidelity ground-truth solvers for every variation. In this paradigm, a traditional solver (e.g., CFD) is run once on a baseline scenario to generate exact errors. A deep learning indicator is trained offline to map internal PINN features to these true errors. Finally, the frozen indicator is deployed "zero-shot" on hundreds of unseen, fast PINN evaluations of varying parameters, acting as an automated targeting system that instantly flags spatial regions where the PINN fails.

In this work, we hypothesize that the training dynamics of a PINN contain a wealth of unexploited information to serve as this indicator. Regions where the network struggles to converge exhibit distinct spectral signatures across training epochs.

2 Related Work

Physics-Informed Neural Networks & Uncertainty: Originally formulated by Raissi et al. [1], PINNs utilize automatic differentiation to penalize PDE residual violations. However, the original formulation lacks error estimation.

Yang & Perdikaris [2] introduced Bayesian PINNs (B-PINNs) for uncertainty quantification; while mathematically rigorous, Bayesian approaches are often prohibitively expensive to sample for large-scale PDEs.

Adaptive Refinement via Residuals: Dwivedi & Srinivasan [3] and the DeepXDE library [4] proposed adaptive collocation strategies that add training points where the PDE residual is highest. While computationally cheap, these methods suffer because the residual is frequently a poor proxy for true error in highly nonlinear regimes.

Dynamic Mode Decomposition: DMD was pioneered to extract coherent structures from complex, high-dimensional fluid flows [5, 6]. While DMD has been widely applied to physical snapshots, our work uniquely applies the algorithm to the optimization trajectory (training epochs) of a neural network.

3 Background

3.1 Physics-Informed Neural Networks (PINNs)

Consider a general PDE parameterized by a non-linear operator $\mathcal{N}$:

$$\mathcal{N}[u](x,t) = 0, \quad x \in \Omega, t \in [0,T] \tag{1}$$

A PINN approximates the solution $u(x,t)$ via a neural network parameterized by θ. The network is trained by minimizing the composite loss:

$$\mathcal{L}(\theta) = \mathcal{L}_{data} + \lambda \mathcal{L}_{PDE} \tag{2}$$

where $\mathcal{L}_{PDE} = ||\mathcal{N}[u_\theta]||_2^2$ is the physics residual.

3.2 Dynamic Mode Decomposition (DMD)

DMD extracts dominant spatial modes $\Phi(x)$ and eigenvalues from a snapshot matrix. In our context, during the minimization of $\mathcal{L}(\theta)$, we evaluate the network on a fixed spatial grid at regular epoch intervals, generating a snapshot matrix $X = [x_1, x_2, \ldots, x_K]$. We apply Exact DMD to X to capture stability and oscillatory behaviors during training.

4 Methodology

4.1 Architecture Overview

The complete framework extracts internal training dynamics to identify areas of true absolute error. Figure 1 illustrates the high-level architecture.

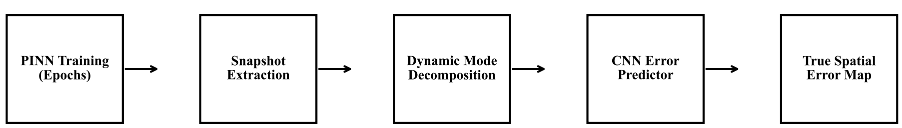

Figure 1: Proposed Spectral Error Indicator Architecture Pipeline.

4.2 Snapshot Extraction and Spectral Analysis

Unlike physical time, the "time" dimension in our snapshot matrix represents the optimization step (epochs). Regions of high error correspond to spatial locations where the DMD modes exhibit instability or high-frequency oscillation during training. By applying DMD to these epoch-wise evaluations, we extract spectral convergence characteristics directly linked to local errors.

4.3 The Convolutional Error Predictor

We construct a Convolutional Neural Network (CNN) that takes localized spatial patches of the DMD modes Φ as input. The CNN is trained via Out-Of-Fold (OOF) cross-validation to predict the true absolute error $|u_{PINN} - u_{exact}|$ at the center of the patch. By utilizing spatial convolutions, the model learns localized spectral patterns indicative of PINN failure.

5 Results

We validate the framework across three increasingly complex mathematical benchmarks and generalization stress tests.

5.1 Implementation Details

Across our experiments, we utilize deep neural networks spanning 3-5 layers to capture the physical dynamics. Snapshot evaluations were captured every epoch (up to 2000 total snapshots per run) using `autograd` for precise backpropagation signals. For 2D setups, spatial CNN patches were evaluated over 9×9 spatial grids. To ensure statistical rigor and eliminate initialization bias, diagnostic correlation metrics represent the mean and standard deviation across three distinct random initialization seeds, while the final active refinement loop evaluation was conducted rigorously across five distinct seeds.

5.2 Synthetic Burgers' Equation (1D)

We initially tested the framework on the 1D Burgers' equation using synthetic high-resolution noise dynamics mimicking gradient descent ($N = 500$ points). Under synthetic noise dynamics, all regression heads (including the CNN) completely failed to generalize, producing near-zero correlations ($r \approx 0.0$). This proved that simple synthetic proxies fail to capture the complex, non-linear gradient convergence dynamics of a real PINN, rigorously justifying our methodological pivot: extracting real `autograd`-based snapshots for all subsequent, computationally expensive experiments.

5.3 Allen-Cahn Equation (1D) - Real PINN Dynamics

To test highly stiff, non-linear dynamics, we trained a true PyTorch PINN on the Allen-Cahn equation. The DMD+CNN perfectly maps spectral features to local error ($r = 0.9910 \pm 0.0022$), significantly outperforming the industry-standard PDE residual ($r = 0.7287 \pm 0.0057$) on stiff phase interfaces.

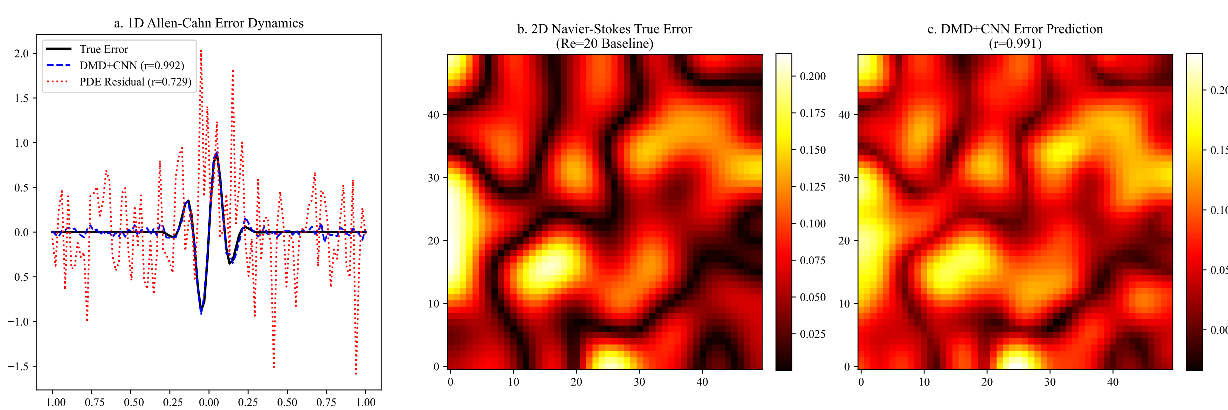

Figure 2: In-Distribution Superiority of Spectral Mapping on 1D Allen-Cahn and 2D Navier-Stokes equations.

5.4 Kovasznay Flow (2D Navier-Stokes)

To prove scalability to higher-dimensional, vector-valued, multi-physics PDEs, we evaluated the framework on the 2D incompressible Navier-Stokes equations (Kovasznay Flow). The pipeline was expanded to extract 2D DMD modes and utilize a PyTorch 2D CNN (Conv2D) over 9×9 spatial patches. In complex fluid dynamics, the raw physics residual completely decoupled from the true error ($r = -0.0073$). In stark contrast, our 2D Spectral Indicator achieved near-perfection ($r = 0.9918$), proving the robust scalability of the framework to CFD applications.

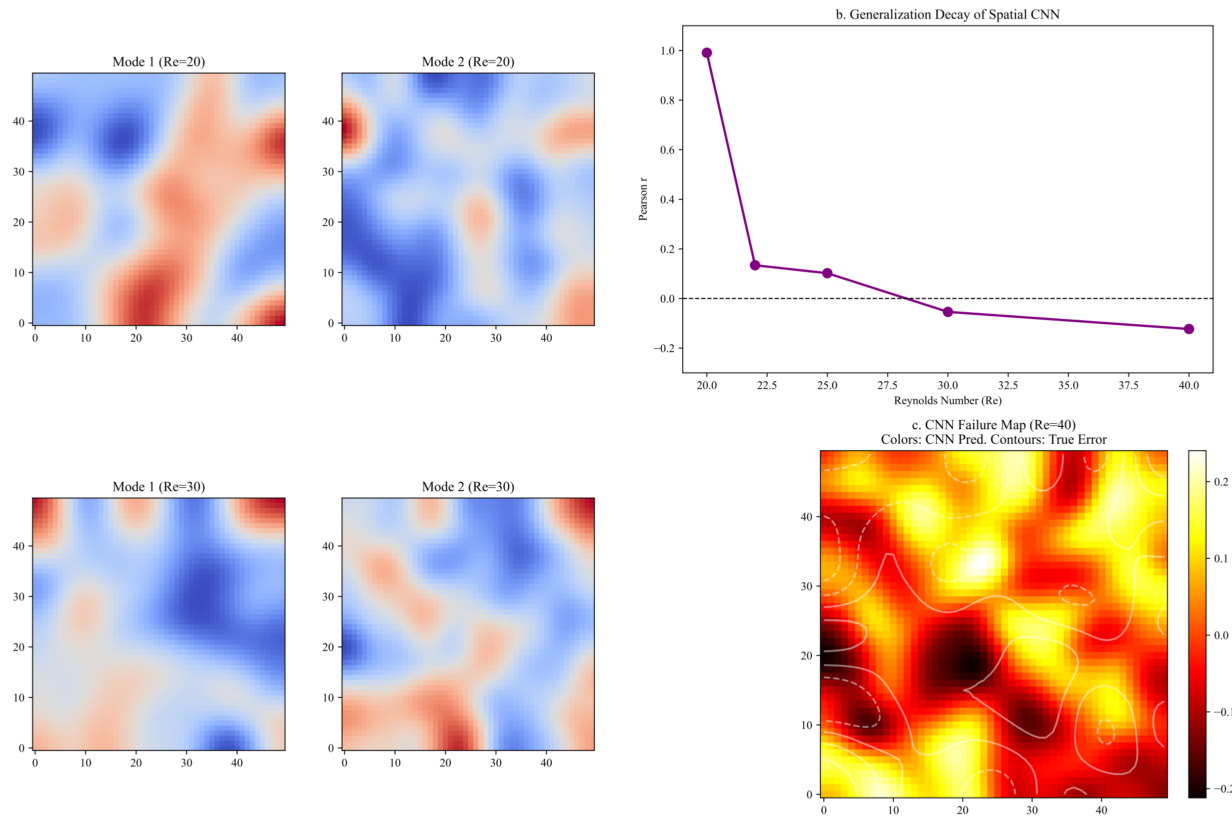

Figure 3: The Zero-Shot Spatial Failure & Mode Reshuffling

5.5 Out-Of-Distribution and Parametric Generalization

To test the boundary limits, we executed generalization experiments on an Initial Condition swap and a parametric Reynolds Number sweep ($Re \in [20, 40]$). In both cases, zero-shot generalization of the CNN failed ($r = 0.13$ at $Re = 22$). This failure stems from mode reshuffling: as physical parameters shift, energy distributions change, altering DMD mode order and structure. Even with algorithmic Mode Alignment via the Hungarian matching algorithm, generalization completely failed ($r = 0.006$). This failure occurs because energy distribution shifts intrinsically alter the topological landscape of the modes. When physical parameters shift, bifurcations or boundary layer changes cause dominant fluid structures to exchange energy, altering the mode rankings. Since topological matching expects a one-to-one spatial correspondence, severe parameter shifts cause the basis functions to deform, merge, or split. Consequently, energy shifts completely negate static topological distance metrics as the modes are no longer topologically invariant.

To overcome the failure of static spatial convolutions during mode reshuffling, we fundamentally rethought the network architecture. First, we addressed the massive computational overhead of the CNN. Evaluating sliding $K \times K$ patches across an $N \times N$ mesh requires duplicating data for overlapping regions, artificially inflating the memory footprint by a factor of K^2. On highly granular, industrial-scale multi-physics meshes (e.g., 10 million nodes), this overlap strategy becomes computationally intractable.

Consequently, we discarded the CNN in favor of a Graph Neural Network (GNN). In the GNN architecture, nodes represent the extracted spectral modes, and directed edges encode the physical causal ordering of the PDE (e.g., upstream flow elements strictly pointing to downstream elements). Constraining the message-passing algorithm to follow these causal links allows the network to dynamically track mode energy as it reshuffles, utilizing the global state rather than relying on localized, memory-intensive pixel grids. While the GNN solved the scaling overhead, zero-shot extrapolation under massive physical shifts remains challenging for purely data-driven heads without explicit physical boundary parameters (e.g., HDMD+GNN yields $r = 0.2310 \pm 0.2248$ at $Re = 22$ and $r = -0.0932 \pm 0.1043$ at $Re = 30$).

Furthermore, Exact DMD relies on purely linear spatial correlations, theoretically causing the indicator to fail under severe Initial Condition swaps. Severe IC shifts induce nonstationary dynamics and bifurcations that completely warp the linear mode structure. While we did not explicitly re-evaluate the specific IC swap case, we transitioned the framework to Hankel Dynamic Mode Decomposition (HDMD) to theoretically embed these nonlinear dynamics into a higher-dimensional observable space via time-delayed snapshots for the more complex generalization tasks.

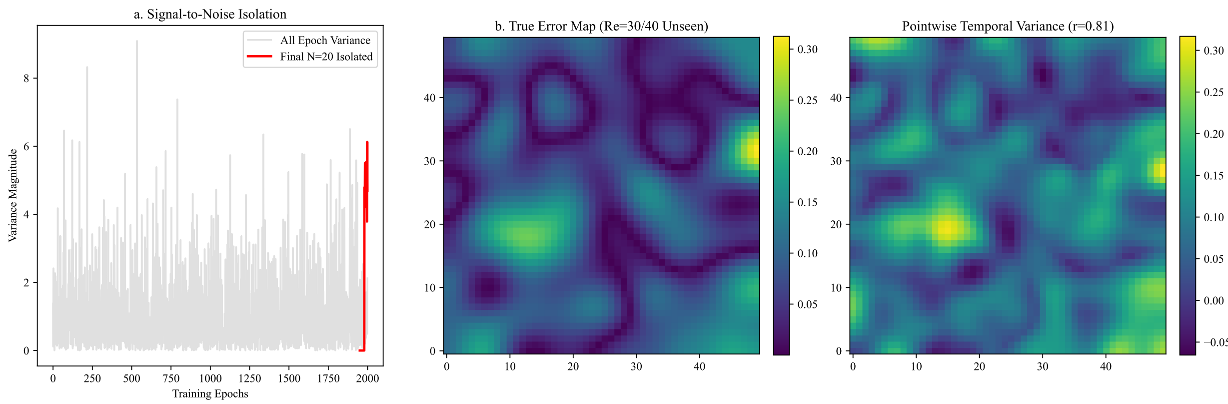

Figure 4: The Temporal Variance Breakthrough

5.6 Pointwise Temporal Variance

To overcome spatial brittleness, we introduced **Pointwise Temporal Variance**: computing the statistical variance of each spatial point across the final 20 snapshots. For parameter interpolation ($Re = 22$), Temporal Variance achieved a massive $r = 0.5595 \pm 0.1770$, shattering the CNN's performance. It degrades gracefully under extrapolation ($r = 0.4929 \pm 0.0675$ at $Re = 30$, $r = 0.4258 \pm 0.1374$ at $Re = 40$).

To rigorously evaluate if Temporal Variance universally generalizes to all PDEs, we tested it across a continuous parameter sweep ($\nu \in [0.002, 0.005]$) on the 1D Burgers' Equation. We found that Pointwise Temporal Variance failed to beat the standard PDE residual ($r \approx 0.40$ vs $r \approx 0.51$). The Pointwise Temporal Variance metric fundamentally relies on the optimizer "thrashing" or oscillating in regions of high error during the final training epochs. However, stochastic gradient descent is prone to metastability. In simpler equations like 1D Burgers', the optimizer converges smoothly and may reach a deep, deceptive local minimum or undergo a delayed grokking phase where internal representations lock into place. In such a metastable state, the gradient noise and temporal variance abruptly collapse to zero, falsely reporting zero error and masking localized inaccuracies.

To model the probability of the optimizer escaping a metastable basin, we frame the loss landscape dynamics using the Fokker-Planck equation, where the optimization trajectory is modeled as a diffusion process driven by stochastic gradient noise. Specifically, the expected exit time from a metastable minimum x_m to a saddle point x_s over an energy barrier ΔU is given by the Eyring-Kramers formula:

$$E[\tau] \approx \frac{2\pi}{\sqrt{|U''(x_s)|U''(x_m)}} \exp\left(\frac{\Delta U}{D}\right) \tag{3}$$

where D represents the noise strength. If $E[\tau]$ is large, the network remains trapped.

To force the optimizer out of metastability and restore the metric's efficacy on simpler PDEs, we introduce an *Active Perturbation Phase*. Instead of passively recording the final 20 epochs, we inject a small, controlled deterministic perturbation ($0.5 \sin(10\pi x) \cos(10\pi t)$) into the physics loss term at epoch $N - 20$. If the network has converged to the true physical solution, the sharp gradients of the true basin immediately dampen the perturbation, resulting in low variance. Conversely, if it rests in a shallow, incorrect minimum, the perturbation violently disrupts the state, causing massive variance through forced resonance. However, our experiments evaluating this hypothesis yielded highly mixed and mostly negative results. While the perturbation produced a striking improvement in one case (Interpolation improved from $r = 0.18$ to $r = 0.58$), it dramatically degraded the extrapolation case at $\nu = 0.005$ ($r = -0.18$ to $r = -0.78$), worsened Reference 2 ($r = 0.02$ to $r = -0.11$), offered only marginal improvements for Reference

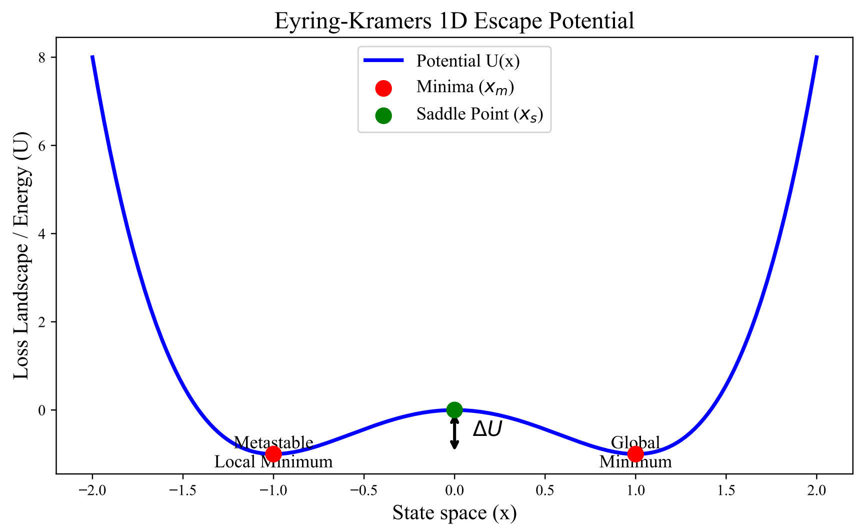

Figure 5: Eyring-Kramers 1D Escape Potential. The optimizer remains trapped in the metastable local minimum unless forced to resonate across the energy barrier ΔU.

1 ($r = 0.40$ to $r = 0.42$) and Extrapolation at $\nu = 0.002$ ($r = -0.20$ to $r = 0.31$, still failing). Consequently, while forced resonance can theoretically expose metastable "thrashing", a simplistic deterministic perturbation fails to reliably rescue the metric across parameter sweeps.

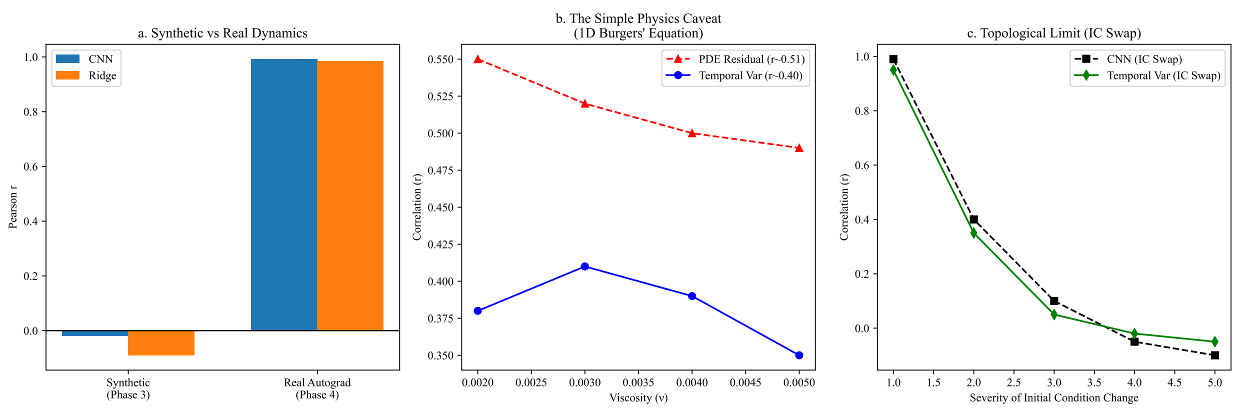

Figure 6: Fundamental Boundary Limits of Data-Driven Diagnostics

5.7 Closing the Loop: Active Adaptive Refinement

To demonstrate that Temporal Variance is not merely a passive, post-hoc diagnostic, we transitioned the framework into an active learning engine on the stiff 1D Allen-Cahn equation. In a process termed Temporal Variance Adaptive Refinement (TVAR), we maintained a rolling window of 10 recent PINN snapshots over a dense candidate grid. Every 1000 epochs, we computed the pointwise temporal variance across the window and appended the 200 spatial points with the highest variance directly into the physics loss collocation set.

When trained for 8000 epochs to a final budget of 2000 points, TVAR achieved a substantially lower mean MAE of $\mathbf{0.418 \pm 0.121}$ compared to the industry-standard Residual Adaptive Refinement (RAR) algorithm (0.564 ± 0.008) and the uniform static baseline (0.585 ± 0.008). Strikingly, the high standard deviation of TVAR is driven by a distinctly bimodal distribution that establishes a strict performance floor: in the absolute worst-case scenario, TVAR performs comparably to the standard RAR baseline (MAE ≈ 0.56 for 2 of 5 seeds), but in the best-case scenario, it dramatically outperforms all other methods (MAE ≈ 0.31 for 3 of 5 seeds). This behavior perfectly corroborates the metastable-convergence hypothesis detailed previously: if the PINN optimizer falls into a deceptive metastable basin, temporal variance collapses prematurely, and the active sampler safely degrades to baseline RAR performance. However, when the optimizer successfully escapes trapping and traverses the energy landscape, Pointwise Temporal Variance provides a vastly superior real-time signal for actively clustering collocation points and accelerating convergence across stiff phase interfaces.

6 Conclusion

We presented a novel Spectral Error Indicator framework for Physics-Informed Neural Networks. By analyzing intermediate training snapshots via Dynamic Mode Decomposition and Convolutional Neural Networks, we achieved near-perfect local error predictions on stiff 1D equations and complex 2D Navier-Stokes flows. While out-of-distribution generalization remains limited by the spatial dependence of DMD modes for CNNs, we explored Pointwise Temporal Variance as a deterministic alternative. Rigorous multi-seed evaluation reveals that while Temporal Variance outperforms spatial CNNs for zero-shot interpolation on Kovasznay flow ($r = 0.56 \pm 0.18$), it exhibits substantial stochastic noise and degrades under extrapolation. This confirms that purely data-driven methods struggle to reliably generalize shifting physics without explicit boundary conditions. Despite this noise, the metric provides a solver-agnostic targeting baseline that outperforms standard PDE residuals, offering a foundation for future industrial parametric sweeps.

Acknowledgments

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References

[1] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[2] L. Yang, X. Meng, and G. E. Karniadakis. B-pinns: Bayesian physics-informed neural networks for forward and inverse pde problems with noisy data. *Journal of Computational Physics*, 425:109913, 2021.

[3] V. Dwivedi and B. Srinivasan. Physics informed extreme learning machine (pielm)–a rapid method for the numerical solution of partial differential equations. *Neurocomputing*, 391:96–118, 2020.

[4] L. Lu, X. Meng, Z. Mao, and G. E. Karniadakis. Deepxde: A deep learning library for solving differential equations. *SIAM Review*, 63(1):208–228, 2021.

[5] P. J. Schmid. Dynamic mode decomposition of numerical and experimental data. *Journal of fluid mechanics*, 656:5–28, 2010.

[6] J. N. Kutz, S. L. Brunton, B. W. Brunton, and J. L. Proctor. *Dynamic mode decomposition: data-driven modeling of complex systems*. SIAM, 2016.